Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Education

Indian Institute of Engineering Science and Technology, Shibpur — Kolkata, India
BTech in Information Technology — *Expected July 2029*

Experience

Software Engineering Intern — Jan 2026 – Mar 2026
Stealth — *Remote*

- Contributed to a payment transaction moderation risk engine, analyzing transactions for policy violations across the BuyMeACoffee (YC W19) platform.
- Reduced false negatives by **1.7%** by replacing a rigid score-based classifier with a **logistic aggregator**, improving recall on edge-case policy violations without degrading precision.

Projects

Spheni – Vector Search Library | *C++, Python, OpenMP* | GitHub — January 2026

- Architected a vector search library from scratch with **Python bindings (pybind11)**, implementing Flat, IVF-Flat, PQ-Flat, and IVF-PQ indexes with **k-means++ clustering** and **inverted list storage**, achieving **12.6× speedup at 97% Recall@10** on 50k SIFT vectors via Voronoi-partitioned candidate scans.
- Implemented **Product Quantization** with residual encoding and **ADC** via precomputed sub-space lookup tables, compressing a 1M × 768-dim Cohere dataset from **2929 MB to 36.6 MB (80×)** at **83.3% Recall@10-in-100**.
- Eliminated **thread contention** in batch query execution via **thread-local Top-K heaps with OpenMP dynamic scheduling**, scaling throughput from **2.4k to 5.7k QPS (2.37×)** and cutting p99 latency from **0.554 ms to 0.287 ms (1.93×)**.
- Built a **multi-sweep benchmark harness** over 1M vectors characterizing recall–latency–memory tradeoffs across nprobe and M.

Rayfloat – CPU Ray Tracer | *C++, OpenMP* | GitHub | Writeup — December 2025

- Architected a physically-based CPU renderer from scratch, implementing Lambertian, Metal, Dielectric, and DiffuseLight materials with an **iterative light transport model**, eliminating **recursion-induced stack pressure** and improving **Branch Target Buffer (BTB) predictability**.
- Diagnosed and eliminated a **41% performance penalty** from **cache-coherence contention** in `std::mt19937` by implementing a **thread-local XorShift32 pRNG**, restoring **near-linear multi-core scaling**.
- Reduced ray-scene intersection complexity from **O(n) to O(log n)** by designing a **Bounding Volume Hierarchy (BVH)** with **median-split construction** and optimized slab AABB tests, enabling scenes with 64,000+ primitives.
- Optimized **dielectric rendering** using **Schlick's Approximation** to numerically estimate the **Fresnel coefficient**, determining the probability of reflection vs refraction based on incident angles.

froginfer – Inference Engine | *C* | GitHub — February 2026

- Implemented a **LLaMA-2 inference engine from scratch in C** with **no external dependencies**, supporting multi-layer transformer decoding with **grouped-query attention (GQA)**, **RoPE positional embeddings**, **RMSNorm**, and **SwiGLU** feed-forward activations.
- Built a **BPE tokenizer** with byte-fallback encoding and greedy merge loop, reproducing SentencePiece-compatible tokenization via iterative bigram scoring and vocabulary lookup over full prompt strings.
- Engineered a **KV-cache** with per-layer key/value buffers across the full context window, enabling $O(1)$ incremental decoding per token and reducing redundant recomputation across autoregressive generation steps.
- Designed **weight layout and memory mapping** via a single contiguous float buffer with pointer arithmetic across embedding, attention, FFN, and classification heads, supporting both **shared and unshared output projection** (tied embeddings).

Skills

Languages: C++, Python, SQL
Systems: OpenMP, Linux, Git
Performance: Intel VTune Profiler, Perf, GDB